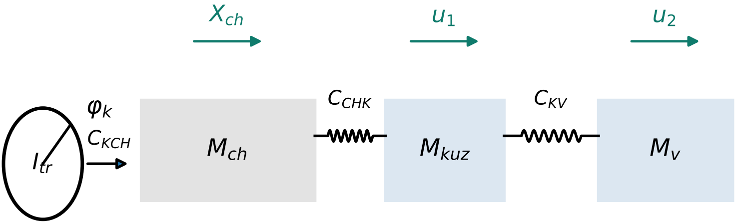

$A_1 = \alpha_d M_\Sigma g \,\mathrm{sgn}\, \dot{X}_{ch}$ $A_2 = \alpha_{ch}(M_{kuz}+M_v) g \,\mathrm{sgn}\, \dot{u}_1$ $A_3 = \alpha_k M_v g \,\mathrm{sgn}\, \dot{u}_2$

$F_{kch} = C_{KCH}(\varphi_k - X_{ch}/R_k)/R_k$ (no viscous damping in the archived worksheet)